Optional Problem.

(1) Give an explicit formula for $J^*(p)$

· Define $J^*(p) = \max_{x \in \mathbb{R}^n} \langle p, x \rangle - J(x)$, $p \in \mathbb{R}^n$,

$$J(x) = \frac{1}{2}\|x\|_2^2 - 1$$

$$J^*(p) = \max_{x \in \mathbb{R}^n} \langle p, x \rangle - \frac{1}{2}\|x\|_2^2 + 1$$

$$= 1 + \max_{x \in \mathbb{R}^n} \left[\langle p, x \rangle - \frac{1}{2}\|x\|_2^2 \right]$$

$$= 1 - \min_{x \in \mathbb{R}^n} \left[\frac{1}{2}\|x\|_2^2 - \langle p, x \rangle \right]$$

Note: $\min \left[\frac{1}{2}\|x\|_2^2 - \langle p, x \rangle \right] = \frac{1}{2}\|p\|_2^2$ at $x = p$

$$J^*(p) = 1 - \frac{1}{2}\|p\|_2^2$$

(2) Give an explicit formula for the proximal point of f.

· Consider $f(y) : y \in \mathbb{R}^n$

$$f(y) = J^*(y) - \langle y, x \rangle$$

$$= \frac{1}{2}\|y\|_2^2 + 1 - \langle y, x \rangle$$

$$\nabla f(y) = y - x$$

And for $\min_y \frac{1}{2}\|y - z\|_2^2 + f(y)$

$F(y) :=$

$$\nabla F(y) = y - z + \nabla f(y)$$

$$= y - z + y - x = 0$$

$$y = \frac{z + x}{2}.$$